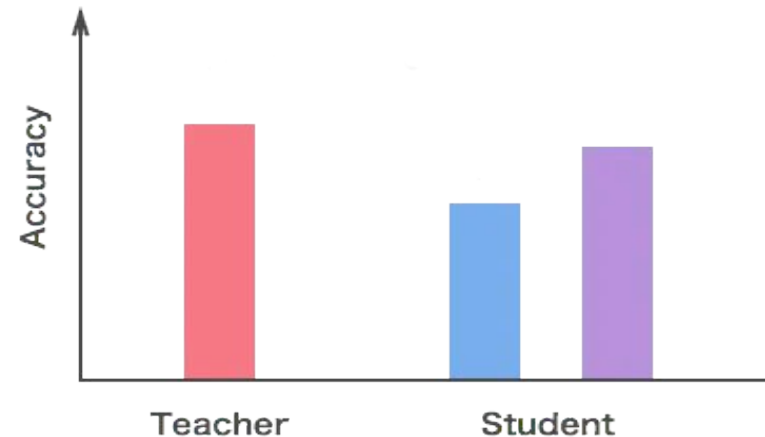
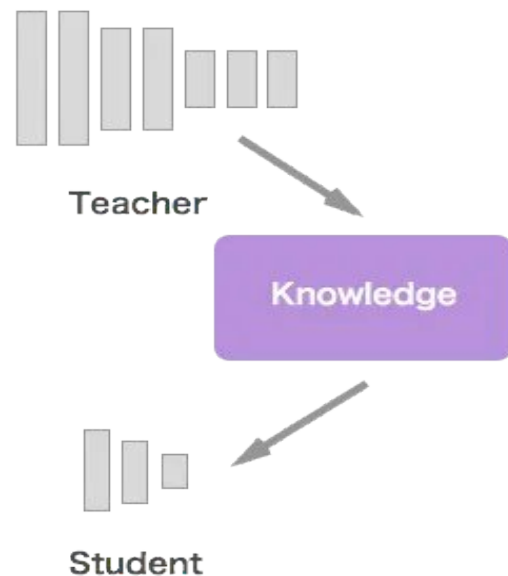


Knowledge Distillation on Deep Neural networks

- Poornima Devi Krishnasamy Karthikeyan -



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Introduction:

Problem:

- Large deep learning models ☾ high accuracy & cost. Real-world deployment ☾ limited memory, smaller & faster models.

Goal :

- Knowledge distillation ☾ lighter models for resource constraint deployment without losing the performance.
 - Transfers knowledge from a large model (Teacher) to a small model (Student).

**Speed is as important as
accuracy**

Dataset Information:

CIFAR-10 :

- 45000 training images
- 5000 validation images
- 10000 test images
- 3 x 32 x 32 images
- 10 classes

airplane



automobile



bird



cat



deer



dog



frog



horse



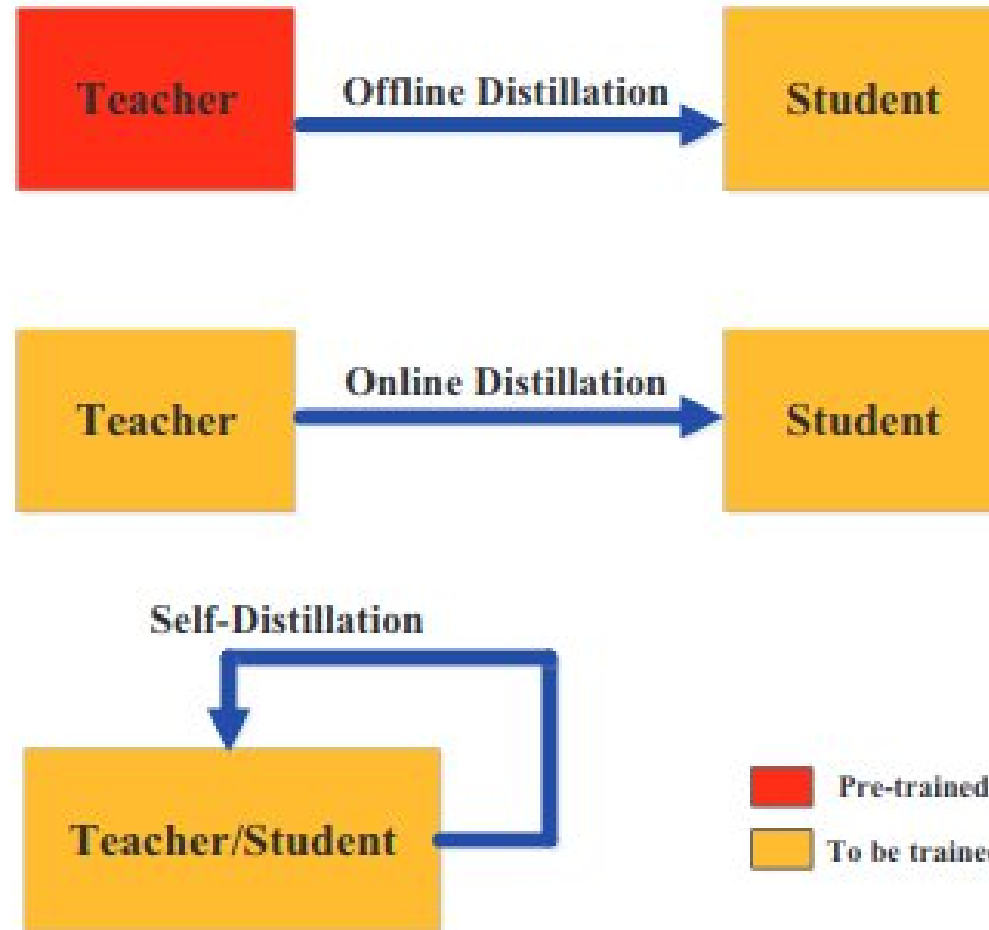
ship



truck

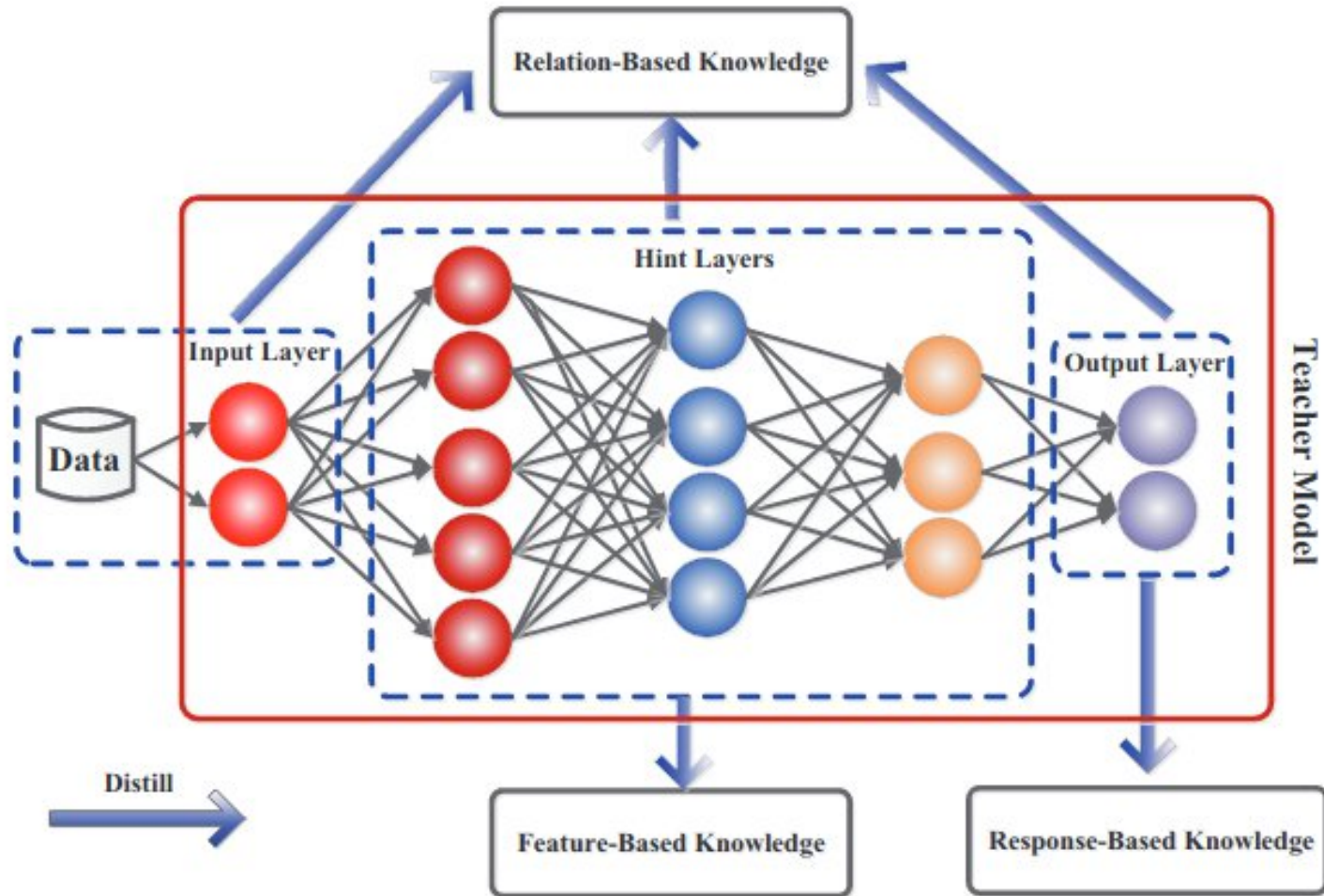


Knowledge distillation schemes:



Offline distillation method is used for the further implementation of KD

Types of Knowledge Distillation:



Methodology : Terms

Softmax :

- raw logits $\hookrightarrow$ probability distribution [0.95, 0.02, 0.01, 0.01, 0.01] $\hookrightarrow$ Not informative beyond the top prediction.

$$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Temperature (T) :

- Softens the probability output from softmax [0.85, 0.05, 0.04, 0.03, 0.01]

$$\frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

- Higher T $\rightarrow$ softer, more informative distribution

Methodology : Terms

Soft targets :

- Teacher's output probabilities [0.85, 0.05, 0.04, 0.03, 0.01] ☾ richer info than just the correct class.

Hard targets :

- One-hot true labels [1, 0, 0, 0] ☾ standard classification loss

Alpha (α):

- Controls the balance between:
 - Distillation loss (soft labels)
 - Classification loss (hard labels)
(how much the student listens to the teacher vs the ground truth)

Methodology - Knowledge Distillation (KD):

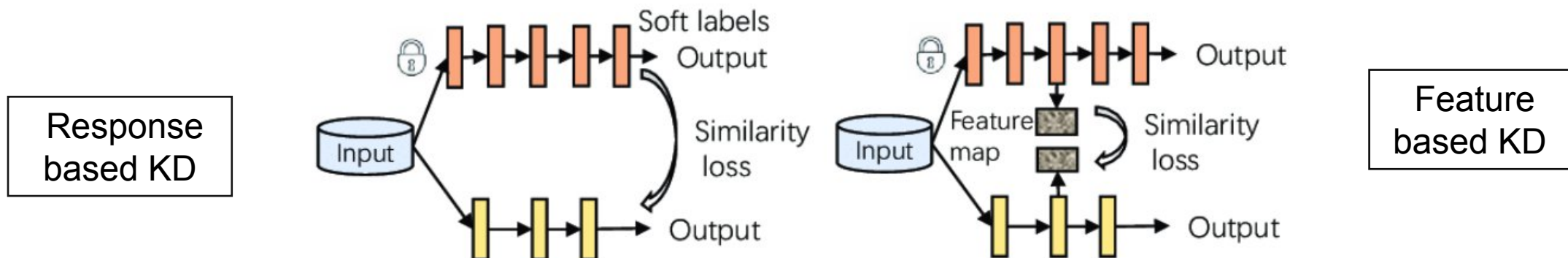
Model architecture:

- Teacher : large pre-trained model (ResNet50) + fine tuned to CIFAR 10
- Student : (ResNet18)

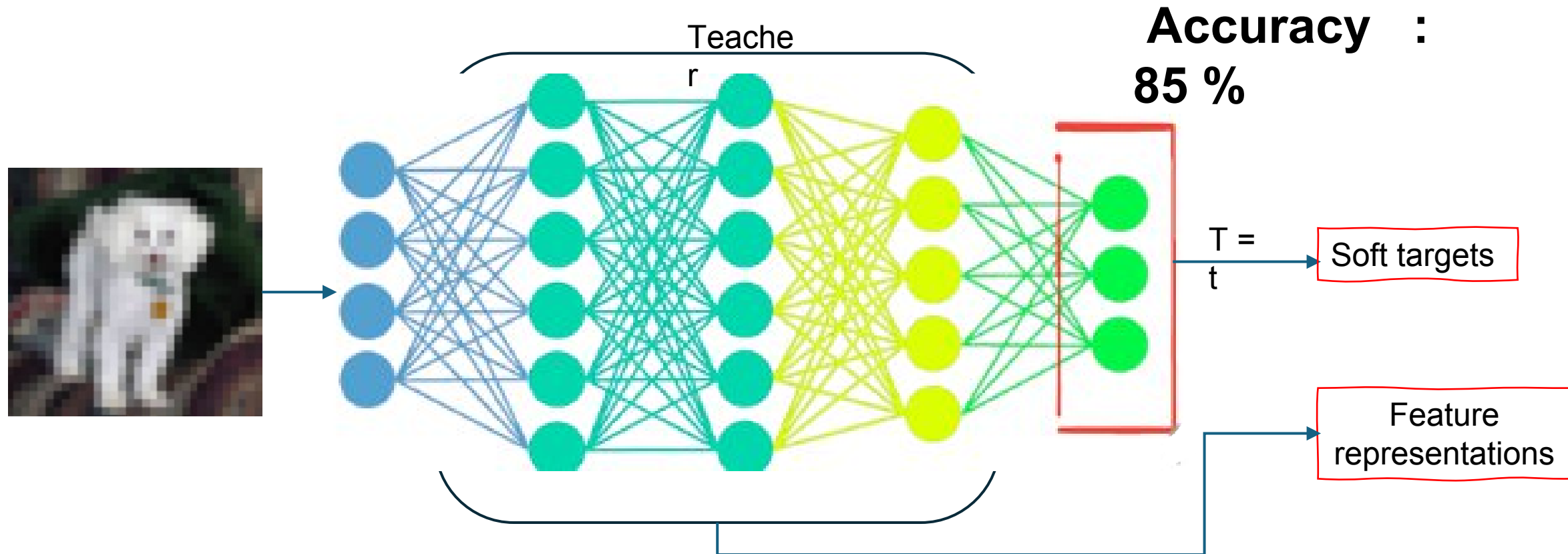
How I trained my student model :

Hybrid Approach = response based KD + feature based KD

- soft outputs ☾ teacher
- Align intermediate features ☾ representation alignment
- Hard labels ☾ student



Implementation – Teacher model



Input $\hookrightarrow$ ResNet50 fine tuned(Teacher) $\hookrightarrow$ extracts Soft Targets + Intermediate Features information

Implementation – Teacher ☾ student

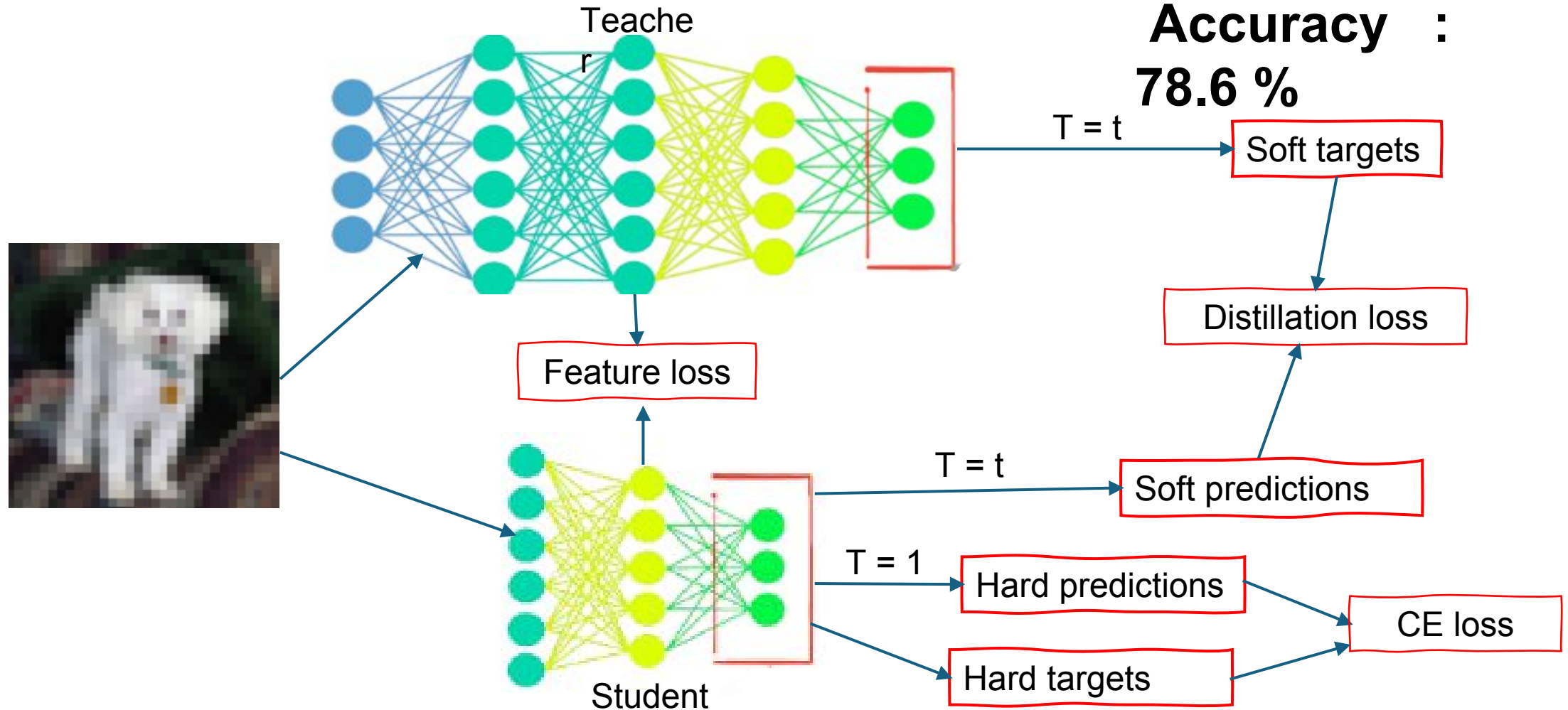
Teacher ☾ student : after projection ☾ [64,128,256,512]

- FeatureProjector ☾ class aligns the teacher's feature maps to the student's feature map.
- 1x1 conv to project teachers features from in_channels (teacher channels) down to out_channels (student channels) , to match the channel dimensions.

```
# define the intermediate feature channels for both teacher and student
student_channels = [64, 128, 256, 512]
teacher_channels = [256, 512, 1024, 2048]

# create projection layers to align teacher's feature maps with student's feature maps
proj_layers = [
    FeatureProjector(in_c, out_c).to(device)
    for in_c, out_c in zip(student_channels, teacher_channels)
]
```

Implementation – student model tuned



Input $\in$ ResNet18 to be trained(student) $\in$ learns from soft targets + intermediate features + hard targets

Implementation – Loss function:

Loss Components:

- Distillation Loss: KL divergence loss
- Classification Loss: Cross entropy loss
- Feature Loss: MSE

Total Loss:

$\mathcal{L} = \alpha \times \text{Distillation Loss} + (1 - \alpha) \times \text{Classification Loss} + 0.1 \times \text{Feature loss}.$

Hyperparameters:

- Temperature: T ☾ controls distribution softness
- Alpha : α ☾ balancing weight for soft and hard losses
- Feature Weight: $\lambda = 0.1$

Implementation – Loss function:

- decreasing MSE $\hookrightarrow$ student model aligning intermediate features with teacher $\hookrightarrow$ improving representation alignment.

```
[(Training student) Epoch 3] Loss = 1.3293 | Feature MSE = 6.4332 |  
New best model saved.  
[(Training student) Epoch 4] Loss = 1.3191 | Feature MSE = 6.4384 |  
New best model saved.  
[(Training student) Epoch 5] Loss = 1.3056 | Feature MSE = 6.2537 |  
New best model saved.  
[(Training student) Epoch 6] Loss = 1.2923 | Feature MSE = 6.1188 |  
New best model saved.  
[(Training student) Epoch 7] Loss = 1.2837 | Feature MSE = 6.0797 |  
New best model saved.  
[(Training student) Epoch 8] Loss = 1.2136 | Feature MSE = 5.8570 |  
New best model saved.  
[(Training student) Epoch 9] Loss = 1.1935 | Feature MSE = 5.8178 |  
New best model saved.  
[(Training student) Epoch 10] Loss = 1.1758 | Feature MSE = 5.7590 |  
[(Training student) Epoch 11] Loss = 1.1606 | Feature MSE = 5.7241 |  
[(Training student) Epoch 12] Loss = 1.1466 | Feature MSE = 5.6620 |  
[(Training student) Epoch 13] Loss = 1.1403 | Feature MSE = 5.6580 |  
[(Training student) Epoch 14] Loss = 1.1290 | Feature MSE = 5.6071 |  
[(Training student) Epoch 15] Loss = 1.1190 | Feature MSE = 5.5648 |  
[(Training student) Epoch 16] Loss = 1.1173 | Feature MSE = 5.5514 |  
[(Training student) Epoch 17] Loss = 1.1063 | Feature MSE = 5.4873 |  
[(Training student) Epoch 18] Loss = 1.1008 | Feature MSE = 5.4574 |  
[(Training student) Epoch 19] Loss = 1.1013 | Feature MSE = 5.4502 |  
[(Training student) Epoch 20] Loss = 1.0752 | Feature MSE = 5.3641 |
```

Implementation -Hyperparameter tuning :

Tuning ☾ two ways

- T , α **fixed** values.
- T , α **scheduling** through exponential decay

Hyperparameter (fixed):

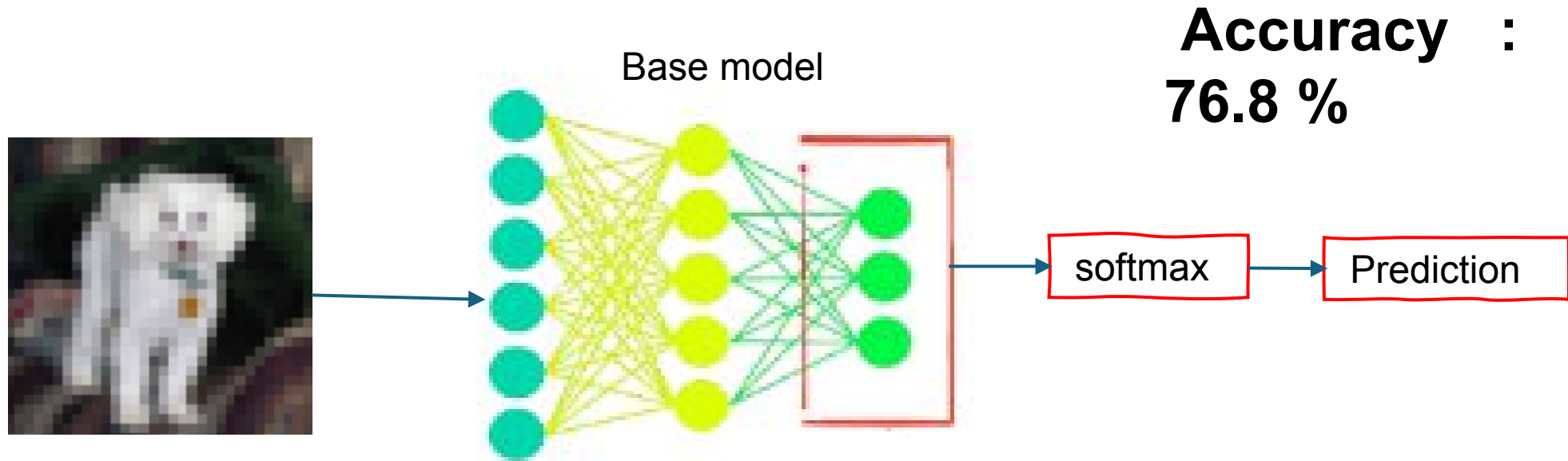
- $T = 0.5$
- $\alpha = 0.7$

Hyperparameter (scheduling) :

- $T = 0.5, 0.3, 0.95$
- $\alpha = 0.8, 0.5, 0.95$

scheduling improved the model accuracy than the fixed values

Implementation – student model without KD



Input $\hookrightarrow$ ResNet18 (trained from scratch) $\hookrightarrow$ predictions (cross entropy loss)

Results :

Model	Architecture	Accuracy (%)	Parameters (M)	Latency (ms)
Teacher	ResNet50	85.11	23.5	8.72
Student distilled (tuned)	ResNet18	78.6	11.18	3.90
Student distilled	ResNet18	76.8	11.18	4.10
Student without KD	ResNet18	75.06	11.18	4.70

Thank you.